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Exercício 5

The image displays two screenshots of the Wireshark network protocol analyzer. The top screenshot shows the initial capture of an HTTP GET request (No. 161) and its corresponding 200 OK response (No. 167). The bottom screenshot provides a detailed view of the reassembled TCP segments and the HTTP response structure.

Top Screenshot: Packet List and Details

No.	Time	Source	Destination	Protocol	Length	Info
161	8.872548458	192.168.1.14	34.223.124.45	HTTP	456	GET / HTTP/1.1
167	9.078382695	34.223.124.45	192.168.1.14	HTTP	911	HTTP/1.1 200 OK (text/html)

Details Panel (Hypertext Transfer Protocol):

- GET / HTTP/1.1\r\n
- Host: neverssl.com\r\n
- Connection: keep-alive\r\n
- Upgrade-Insecure-Requests: 1\r\n
- User-Agent: Mozilla/5.0 (X11; Linux x86_64) AppleWebKit/537.36 (KHTML; like Gecko) Chrome/14.0.0.0 Safari/537.36
- Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,*/*;q=0.8
- Accept-Language: pt-BR,pt;q=0.9\r\n
- Accept-Encoding: gzip, deflate\r\n
- [Full request URL: http://neverssl.com/]

Bottom Screenshot: Reassembled TCP Segments and HTTP Response Structure

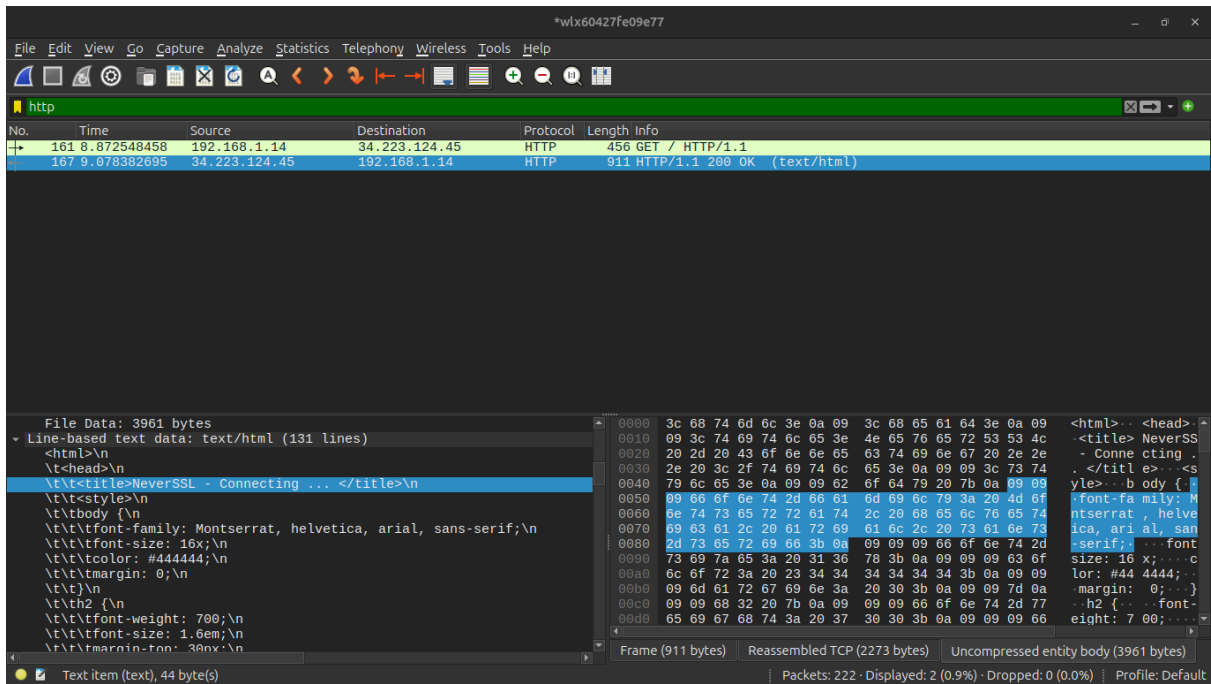
[2 Reassembled TCP Segments (2273 bytes): #165(1428), #167(845)]

Hypertext Transfer Protocol

- HTTP/1.1 200 OK\r\n
- Date: Sat, 06 Jun 2020 04:35:21 GMT\r\n
- Server: Apache/2.4.66 (Ubuntu)\r\n
- Upgrade: h2,h2c\r\n
- Connection: Upgrade, Keep-Alive\r\n
- Last-Modified: Wed, 29 Jun 2022 00:23:33 GMT\r\n
- ETag: "f79-5e28b29d38e93-gzip"\r\n
- Accept-Ranges: bytes\r\n
- Vary: Accept-Encoding\r\n
- Content-Encoding: gzip\r\n
- Content-Length: 1980\r\n
- Keep-Alive: timeout=5, max=100\r\n
- Content-Type: text/html; charset=UTF-8\r\n
- \r\n

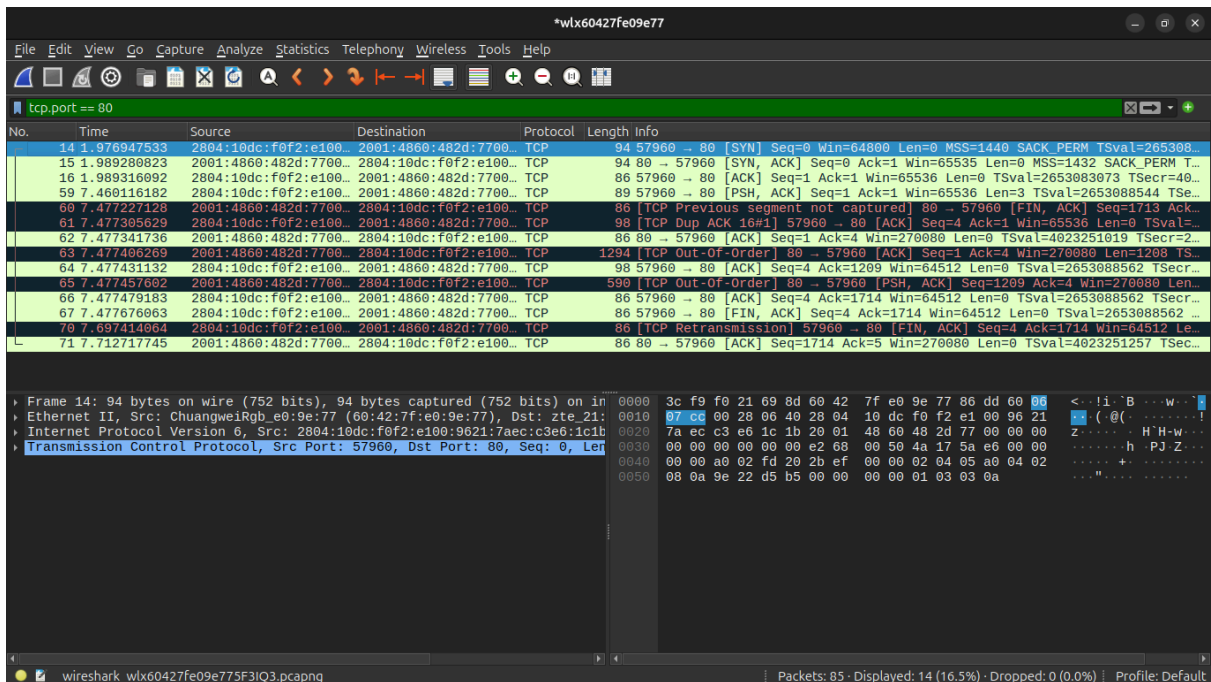
Frame (911 bytes) | Reassembled TCP (2273 bytes) | Uncompressed entity body (3961 bytes)

Packets: 222 · Displayed: 2 (0.9%) · Dropped: 0 (0.0%) · Profile: Default



A requisição utilizou o metodo GET para o Host neverssl.com pelo navegador Mozilla. A resposta do servidor foi um HTTP 200 OK na data 6 de jun de 2026, o server usa Apache, a ultima modificação foi em 29 de junho de 2022 o content-Type é text/html indicando que é uma pagina web. Logo abaixo das linhas de cabeçalho, tem uma seção chamada "Line-based text data: text/html" onde é possível ver o html dessa página.

Exercício 6



*wlx60427fe09e77

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tcp.port == 80

No.	Time	Source	Destination	Protocol	Length	Info
14	1.976947533	2804:10dc:f0f2:e100...	2001:4860:482d:7700...	TCP	94	57960 → 80 [SYN] Seq=0 Win=64800 Len=0 MSS=1440 SACK_PERM TSval=265308...
15	1.989280823	2001:4860:482d:7700...	2804:10dc:f0f2:e100...	TCP	94	80 → 57960 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1432 SACK_PERM T...
16	1.989316092	2804:10dc:f0f2:e100...	2001:4860:482d:7700...	TCP	86	57960 → 80 [ACK] Seq=1 Ack=1 Win=65536 Len=0 TSval=2653083073 TSecr=40...
59	7.460116182	2804:10dc:f0f2:e100...	2001:4860:482d:7700...	TCP	89	57960 → 80 [PSH, ACK] Seq=1 Ack=1 Win=65536 Len=3 TSval=2653088544 TSe...
60	7.477227128	2001:4860:482d:7700...	2804:10dc:f0f2:e100...	TCP	86	[TCP Previous segment not captured] 80 → 57960 [FIN, ACK] Seq=1713 Ack...
61	7.477305629	2804:10dc:f0f2:e100...	2001:4860:482d:7700...	TCP	98	[TCP Dup ACK 16#1] 57960 → 80 [ACK] Seq=4 Ack=1 Win=65536 Len=0 TSval=...
62	7.477341736	2001:4860:482d:7700...	2804:10dc:f0f2:e100...	TCP	86	80 → 57960 [ACK] Seq=1 Ack=4 Win=270080 Len=0 TSval=4023251019 TSecr=2...
63	7.477406269	2001:4860:482d:7700...	2804:10dc:f0f2:e100...	TCP	1294	[TCP Out-Of-Order] 80 → 57960 [ACK] Seq=1 Ack=4 Win=270080 Len=1208 TS...
64	7.477431132	2804:10dc:f0f2:e100...	2001:4860:482d:7700...	TCP	98	57960 → 80 [ACK] Seq=4 Ack=1209 Win=64512 Len=0 TSval=2653088562 TSecr...
65	7.477457602	2001:4860:482d:7700...	2804:10dc:f0f2:e100...	TCP	590	[TCP Out-Of-Order] 80 → 57960 [PSH, ACK] Seq=1209 Ack=4 Win=270080 Len...
66	7.477479183	2804:10dc:f0f2:e100...	2001:4860:482d:7700...	TCP	86	57960 → 80 [ACK] Seq=4 Ack=1714 Win=64512 Len=0 TSval=2653088562 TSecr...
67	7.477676063	2804:10dc:f0f2:e100...	2001:4860:482d:7700...	TCP	86	57960 → 80 [FIN, ACK] Seq=4 Ack=1714 Win=64512 Len=0 TSval=2653088562 ...
70	7.697414064	2804:10dc:f0f2:e100...	2001:4860:482d:7700...	TCP	86	[TCP Retransmission] 57960 → 80 [FIN, ACK] Seq=4 Ack=1714 Win=64512 Le...
71	7.712717745	2001:4860:482d:7700...	2804:10dc:f0f2:e100...	TCP	86	80 → 57960 [ACK] Seq=1714 Ack=5 Win=270080 Len=0 TSval=4023251257 TSec...

Acknowledgment Number: 0
Acknowledgment number (raw): 0
1010 = Header Length: 40 bytes (10)
Flags: 0x002 (SYN)
0000 = Reserved: Not set
...0 = Accurate ECN: Not set
....0... = Congestion Window Reduced: Not set
....0... = ECN-Echo: Not set
....0... = Urgent: Not set
....0... = Acknowledgment: Not set
....0... = Push: Not set
....0... = Reset: Not set
....1... = Syn: Set
....0... = Fin: Not set
[TCP Flags:S.]
Window: 64800
[Calculated window size: 64800]

0000 3c f9 f0 21 69 8d 60 42 7f e0 9e 77 86 dd 60 00 <...i.B...w...!
0010 07 c0 00 28 06 40 28 04 10 dc f0 f2 e1 00 96 21 w...y...H'H-w...
0020 7a ec c3 e6 1c 1b 20 01 48 60 48 2d 77 00 00 00 z...P...h'gXJ...
0030 00 00 00 00 00 00 e2 68 00 50 4a 17 5a e6 00 00 Z...P...h'gXJ...
0040 00 00 a0 02 fd 20 2b ef 00 00 02 04 05 a0 04 02 Z...P...h'gXJ...
0050 08 0a 9e 22 d5 b5 00 00 00 00 01 03 03 0a Z...P...h'gXJ...

wireshark_wlx60427fe09e775f3IQ3.pcapng Packets: 85 · Displayed: 14 (16.5%) · Dropped: 0 (0.0%) · Profile: Default

*wlx60427fe09e77

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tcp.port == 80

No.	Time	Source	Destination	Protocol	Length	Info
14	1.976947533	2804:10dc:f0f2:e100...	2001:4860:482d:7700...	TCP	94	57960 → 80 [SYN] Seq=0 Win=64800 Len=0 MSS=1440 SACK_PERM TSval=265308...
15	1.989280823	2001:4860:482d:7700...	2804:10dc:f0f2:e100...	TCP	94	80 → 57960 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1432 SACK_PERM T...
16	1.989316092	2804:10dc:f0f2:e100...	2001:4860:482d:7700...	TCP	86	57960 → 80 [ACK] Seq=1 Ack=1 Win=65536 Len=0 TSval=2653083073 TSecr=40...
59	7.460116182	2804:10dc:f0f2:e100...	2001:4860:482d:7700...	TCP	89	57960 → 80 [PSH, ACK] Seq=1 Ack=1 Win=65536 Len=3 TSval=2653088544 TSe...
60	7.477227128	2001:4860:482d:7700...	2804:10dc:f0f2:e100...	TCP	86	[TCP Previous segment not captured] 80 → 57960 [FIN, ACK] Seq=1713 Ack...
61	7.477305629	2804:10dc:f0f2:e100...	2001:4860:482d:7700...	TCP	98	[TCP Dup ACK 16#1] 57960 → 80 [ACK] Seq=4 Ack=1 Win=65536 Len=0 TSval=...
62	7.477341736	2001:4860:482d:7700...	2804:10dc:f0f2:e100...	TCP	86	80 → 57960 [ACK] Seq=1 Ack=4 Win=270080 Len=0 TSval=4023251019 TSecr=2...
63	7.477406269	2001:4860:482d:7700...	2804:10dc:f0f2:e100...	TCP	1294	[TCP Out-Of-Order] 80 → 57960 [ACK] Seq=1 Ack=4 Win=270080 Len=1208 TS...
64	7.477431132	2804:10dc:f0f2:e100...	2001:4860:482d:7700...	TCP	98	57960 → 80 [ACK] Seq=4 Ack=1209 Win=64512 Len=0 TSval=2653088562 TSecr...
65	7.477457602	2001:4860:482d:7700...	2804:10dc:f0f2:e100...	TCP	590	[TCP Out-Of-Order] 80 → 57960 [PSH, ACK] Seq=1209 Ack=4 Win=270080 Len...
66	7.477479183	2804:10dc:f0f2:e100...	2001:4860:482d:7700...	TCP	86	57960 → 80 [ACK] Seq=4 Ack=1714 Win=64512 Len=0 TSval=2653088562 TSecr...
67	7.477676063	2804:10dc:f0f2:e100...	2001:4860:482d:7700...	TCP	86	57960 → 80 [FIN, ACK] Seq=4 Ack=1714 Win=64512 Len=0 TSval=2653088562 ...
70	7.697414064	2804:10dc:f0f2:e100...	2001:4860:482d:7700...	TCP	86	[TCP Retransmission] 57960 → 80 [FIN, ACK] Seq=4 Ack=1714 Win=64512 Le...
71	7.712717745	2001:4860:482d:7700...	2804:10dc:f0f2:e100...	TCP	86	80 → 57960 [ACK] Seq=1714 Ack=5 Win=270080 Len=0 TSval=4023251257 TSec...

Acknowledgment Number: 1 (relative ack number)
Acknowledgment number (raw): 124304583
1010 = Header Length: 40 bytes (10)
Flags: 0x012 (SYN, ACK)
0000 = Reserved: Not set
...0 = Accurate ECN: Not set
....0... = Congestion Window Reduced: Not set
....0... = ECN-Echo: Not set
....0... = Urgent: Not set
....1... = Acknowledgment: Set
....0... = Push: Not set
....0... = Reset: Not set
....1... = Syn: Set
....0... = Fin: Not set
[TCP Flags:A..S.]
Window: 65535
[Calculated window size: 65535]

0000 60 42 7f e0 9e 77 3c f9 f0 21 69 8d 86 dd 68 07 B...w...i...h...
0010 d9 77 00 28 06 79 20 01 48 60 48 2d 77 00 00 00 w...y...H'H-w...
0020 00 00 00 00 00 00 28 04 10 dc f0 f2 e1 00 96 21(.....!
0030 7a ec c3 e6 1c 1b 00 50 e2 68 27 27 67 58 4a 17 z...P...h'gXJ...
0040 5a e7 a0 12 ff ff cf dc 00 00 02 04 05 98 04 02 Z...P...h'gXJ...
0050 08 0a ef cd da de 9e 22 d5 b5 01 03 03 08 Z...P...h'gXJ...

wireshark_wlx60427fe09e775f3IQ3.pcapng Packets: 85 · Displayed: 14 (16.5%) · Dropped: 0 (0.0%) · Profile: Default

*wlx60427fe09e77

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tcp.port == 80

No.	Time	Source	Destination	Protocol	Length	Info
14	1.976947533	2804:10dc:f0f2:e100...	2001:4860:482d:7700...	TCP	94	57960 → 80 [SYN] Seq=0 Win=64800 Len=0 MSS=1440 SACK_PERM TSval=265308...
15	1.989280823	2001:4860:482d:7700...	2804:10dc:f0f2:e100...	TCP	94	80 → 57960 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1432 SACK_PERM T...
16	1.989316092	2804:10dc:f0f2:e100...	2001:4860:482d:7700...	TCP	86	57960 → 80 [ACK] Seq=1 Ack=1 Win=65536 Len=0 TSval=2653083073 TSecr=40...
59	7.460116182	2804:10dc:f0f2:e100...	2001:4860:482d:7700...	TCP	89	57960 → 80 [PSH, ACK] Seq=1 Ack=1 Win=65536 Len=3 TSval=2653088544 TSe...
60	7.477227128	2001:4860:482d:7700...	2804:10dc:f0f2:e100...	TCP	86	[TCP Previous segment not captured] 80 → 57960 [FIN, ACK] Seq=1713 Ack...
61	7.477305629	2804:10dc:f0f2:e100...	2001:4860:482d:7700...	TCP	98	[TCP Dup ACK 16#1] 57960 → 80 [ACK] Seq=4 Ack=1 Win=65536 Len=0 TSval=...
62	7.477341736	2001:4860:482d:7700...	2804:10dc:f0f2:e100...	TCP	86	80 → 57960 [ACK] Seq=1 Ack=4 Win=270080 Len=0 TSval=4023251019 TSecr=2...
63	7.477406269	2001:4860:482d:7700...	2804:10dc:f0f2:e100...	TCP	1294	[TCP Out-Of-Order] 80 → 57960 [ACK] Seq=1 Ack=4 Win=270080 Len=1208 TS...
64	7.477431132	2804:10dc:f0f2:e100...	2001:4860:482d:7700...	TCP	98	57960 → 80 [ACK] Seq=4 Ack=1209 Win=64512 Len=0 TSval=2653088562 TSecr...
65	7.477457602	2001:4860:482d:7700...	2804:10dc:f0f2:e100...	TCP	590	[TCP Out-Of-Order] 80 → 57960 [PSH, ACK] Seq=1209 Ack=4 Win=270080 Len...
66	7.477479183	2804:10dc:f0f2:e100...	2001:4860:482d:7700...	TCP	86	57960 → 80 [ACK] Seq=4 Ack=1714 Win=64512 Len=0 TSval=2653088562 TSecr...
67	7.477676063	2804:10dc:f0f2:e100...	2001:4860:482d:7700...	TCP	86	57960 → 80 [FIN, ACK] Seq=4 Ack=1714 Win=64512 Len=0 TSval=2653088562 ...
70	7.697414064	2804:10dc:f0f2:e100...	2001:4860:482d:7700...	TCP	86	[TCP Retransmission] 57960 → 80 [FIN, ACK] Seq=4 Ack=1714 Win=64512 Le...
71	7.712717745	2001:4860:482d:7700...	2804:10dc:f0f2:e100...	TCP	86	80 → 57960 [ACK] Seq=1714 Ack=5 Win=270080 Len=0 TSval=4023251257 TSec...

Acknowledgment number (raw): 656895498
1000 = Header Length: 32 bytes (8)
Flags: 0x011 (FIN, ACK)
0000 = Reserved: Not set
...0 = Accurate ECN: Not set
...0 = Congestion Window Reduced: Not set
...0 = ECN-Echo: Not set
...0 = Urgent: Not set
...1 = Acknowledgment: Set
...0 = Push: Not set
...0 = Reset: Not set
...0 = Syn: Not set
...1 = Fin: Set
[Expert Info (Chat/Sequence): Connection finish (FIN)]
[TCP Flags:A..F]
[Expert Info (Note/Sequence): This frame undergoes the connection
Window: 63

Reset (tcp.flags.reset), 1 byte(s)

Packets: 85 · Displayed: 14 (16.5%) · Dropped: 0 (0.0%) · Profile: Default

*wlx60427fe09e77

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tcp.port == 80

No.	Time	Source	Destination	Protocol	Length	Info
14	1.976947533	2804:10dc:f0f2:e100...	2001:4860:482d:7700...	TCP	94	57960 → 80 [SYN] Seq=0 Win=64800 Len=0 MSS=1440 SACK_PERM TSval=265308...
15	1.989280823	2001:4860:482d:7700...	2804:10dc:f0f2:e100...	TCP	94	80 → 57960 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1432 SACK_PERM T...
16	1.989316092	2804:10dc:f0f2:e100...	2001:4860:482d:7700...	TCP	86	57960 → 80 [ACK] Seq=1 Ack=1 Win=65536 Len=0 TSval=2653083073 TSecr=40...
59	7.460116182	2804:10dc:f0f2:e100...	2001:4860:482d:7700...	TCP	89	57960 → 80 [PSH, ACK] Seq=1 Ack=1 Win=65536 Len=3 TSval=2653088544 TSe...
60	7.477227128	2001:4860:482d:7700...	2804:10dc:f0f2:e100...	TCP	86	[TCP Previous segment not captured] 80 → 57960 [FIN, ACK] Seq=1713 Ack...
61	7.477305629	2804:10dc:f0f2:e100...	2001:4860:482d:7700...	TCP	98	[TCP Dup ACK 16#1] 57960 → 80 [ACK] Seq=4 Ack=1 Win=65536 Len=0 TSval=...
62	7.477341736	2001:4860:482d:7700...	2804:10dc:f0f2:e100...	TCP	86	80 → 57960 [ACK] Seq=1 Ack=4 Win=270080 Len=0 TSval=4023251019 TSecr=2...
63	7.477406269	2001:4860:482d:7700...	2804:10dc:f0f2:e100...	TCP	1294	[TCP Out-Of-Order] 80 → 57960 [ACK] Seq=1 Ack=4 Win=270080 Len=1208 TS...
64	7.477431132	2804:10dc:f0f2:e100...	2001:4860:482d:7700...	TCP	98	57960 → 80 [ACK] Seq=4 Ack=1209 Win=64512 Len=0 TSval=2653088562 TSecr...
65	7.477457602	2001:4860:482d:7700...	2804:10dc:f0f2:e100...	TCP	590	[TCP Out-Of-Order] 80 → 57960 [PSH, ACK] Seq=1209 Ack=4 Win=270080 Len...
66	7.477479183	2804:10dc:f0f2:e100...	2001:4860:482d:7700...	TCP	86	57960 → 80 [ACK] Seq=4 Ack=1714 Win=64512 Len=0 TSval=2653088562 TSecr...
67	7.477676063	2804:10dc:f0f2:e100...	2001:4860:482d:7700...	TCP	86	57960 → 80 [FIN, ACK] Seq=4 Ack=1714 Win=64512 Len=0 TSval=2653088562 ...
70	7.697414064	2804:10dc:f0f2:e100...	2001:4860:482d:7700...	TCP	86	[TCP Retransmission] 57960 → 80 [FIN, ACK] Seq=4 Ack=1714 Win=64512 Le...
71	7.712717745	2001:4860:482d:7700...	2804:10dc:f0f2:e100...	TCP	86	80 → 57960 [ACK] Seq=1714 Ack=5 Win=270080 Len=0 TSval=4023251257 TSec...

Acknowledgment number (raw): 1243944597
1000 = Header Length: 32 bytes (8)
Flags: 0x010 (ACK)
0000 = Reserved: Not set
...0 = Accurate ECN: Not set
...0 = Congestion Window Reduced: Not set
...0 = ECN-Echo: Not set
...0 = Urgent: Not set
...1 = Acknowledgment: Set
...0 = Push: Not set
...0 = Reset: Not set
...0 = Syn: Not set
...0 = Fin: Not set
[TCP Flags:A....]
Window: 1055
[Calculated window size: 270080]
[Window size scaling factor: 256]

Acknowledgment (tcp.flags.ack), 1 byte(s)

Packets: 85 · Displayed: 14 (16.5%) · Dropped: 0 (0.0%) · Profile: Default

Na primeira linha é possível ver o envio do flag SYN para fazer a conexão com a porta 80 do servidor da Google, expandindo as flags podemos ver que a flag Syn está marcada com 1. Na segunda linha é possível ver que o servidor respondeu com as flags Syn e Acknowledgment marcados com 1 e logo em seguida é enviado uma flag ACK do meu computador para o servidor da Google, assim estabelecendo a conexão entre os dois. No encerramento da conexão onde digitei uma tecla qualquer e apertei enter, assim o meu computador enviou com as flags FIN e ACK marcados com 1 e esse pacote foi retransmitindo logo abaixo e então teve a resposta do servidor da Google com o flag ACK marcado com 1.

Exercício 7

*wtx60427fe09e77

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dns

No.	Time	Source	Destination	Protocol	Length	Info
9	2.419494092	2804:10dc:f0f2:e100...	2804:10dc:8100:22::...	DNS	108	Standard query 0xd557 A www.wikipedia.org OPT
10	2.427128390	2804:10dc:8100:22::...	2804:10dc:f0f2:e100...	DNS	153	Standard query response 0xd557 A www.wikipedia.org CNAME dyna.wikimedi...
11	2.428069983	2804:10dc:f0f2:e100...	2804:10dc:8100:22::...	DNS	109	Standard query 0xdde3 AAAA dyna.wikimedia.org OPT
12	2.432695888	2804:10dc:8100:22::...	2804:10dc:f0f2:e100...	DNS	137	Standard query response 0xdde3 AAAA dyna.wikimedia.org AAAA 2a02:ec80:...

User Datagram Protocol, Src Port: 35965, Dst Port: 53

Domain Name System (query)

Transaction ID: 0xd557

Flags: 0x0100 Standard query

Questions: 1

Answer RRs: 0

Authority RRs: 0

Additional RRs: 1

Queries

- www.wikipedia.org: type A, class IN
 - Name: www.wikipedia.org
 - [Name Length: 17]
 - [Label Count: 3]
 - Type: A (1) (Host Address)
 - Class: IN (0x0001)

Additional records

[Response In: 10]

Domain Name System: Protocol

Packets: 43 · Displayed: 4 (9.3%) · Dropped: 0 (0.0%) · Profile: Default

*wtx60427fe09e77

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

dns

No.	Time	Source	Destination	Protocol	Length	Info
9	2.419494092	2804:10dc:f0f2:e100...	2804:10dc:8100:22::...	DNS	108	Standard query 0xd557 A www.wikipedia.org OPT
10	2.427128390	2804:10dc:8100:22::...	2804:10dc:f0f2:e100...	DNS	153	Standard query response 0xd557 A www.wikipedia.org CNAME dyna.wikimedi...
11	2.428069983	2804:10dc:f0f2:e100...	2804:10dc:8100:22::...	DNS	109	Standard query 0xdde3 AAAA dyna.wikimedia.org OPT
12	2.432695888	2804:10dc:8100:22::...	2804:10dc:f0f2:e100...	DNS	137	Standard query response 0xdde3 AAAA dyna.wikimedia.org AAAA 2a02:ec80:...

Questions: 1

Answer RRs: 2

Authority RRs: 0

Additional RRs: 1

Queries

- www.wikipedia.org: type A, class IN
 - Name: www.wikipedia.org
 - [Name Length: 17]
 - [Label Count: 3]
 - Type: A (1) (Host Address)
 - Class: IN (0x0001)

Answers

- www.wikipedia.org: type CNAME, class IN, cname dyna.wikimedia.org
- dyna.wikimedia.org: type A, class IN, addr 195.200.68.224

Additional records

[Request In: 9]

[Time: 0.007634298 seconds]

Domain Name System: Protocol

Packets: 43 · Displayed: 4 (9.3%) · Dropped: 0 (0.0%) · Profile: Default

*wlx60427fe09e77

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

dns

No.	Time	Source	Destination	Protocol	Length	Info
9	2.419494092	2804:10dc:f0f2:e100...	2804:10dc:8100:22::...	DNS	108	Standard query 0xd557 A www.wikipedia.org OPT
10	2.427128390	2804:10dc:8100:22::...	2804:10dc:f0f2:e100...	DNS	153	Standard query response 0xd557 A www.wikipedia.org CNAME dyna.wikimedi...
11	2.428069983	2804:10dc:f0f2:e100...	2804:10dc:8100:22::...	DNS	109	Standard query 0xdd3 AAAA dyna.wikimedia.org OPT
12	2.432695888	2804:10dc:8100:22::...	2804:10dc:f0f2:e100...	DNS	137	Standard query response 0xdd3 AAAA dyna.wikimedia.org AAAA 2a02:ec80:...

Domain Name System (query)
Transaction ID: 0xdd3
Flags: 0x0100 Standard query
Questions: 1
Answer RRs: 0
Authority RRs: 0
Additional RRs: 1
Queries
dyna.wikimedia.org: type AAAA, class IN
Name: dyna.wikimedia.org
[Name Length: 18]
[Label Count: 3]
Type: AAAA (28) (IP6 Address)
Class: IN (0x0001)
Additional records
Root: type OPT
[Response In: 12]

Domain Name System: Protocol | Packets: 43 · Displayed: 4 (9.3%) · Dropped: 0 (0.0%) | Profile: Default

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dns

No.	Time	Source	Destination	Protocol	Length	Info
9	2.419494092	2804:10dc:f0f2:e100...	2804:10dc:8100:22::...	DNS	108	Standard query 0xd557 A www.wikipedia.org OPT
10	2.427128390	2804:10dc:8100:22::...	2804:10dc:f0f2:e100...	DNS	153	Standard query response 0xd557 A www.wikipedia.org CNAME dyna.wikimedi...
11	2.428069983	2804:10dc:f0f2:e100...	2804:10dc:8100:22::...	DNS	109	Standard query 0xdd3 AAAA dyna.wikimedia.org OPT
12	2.432695888	2804:10dc:8100:22::...	2804:10dc:f0f2:e100...	DNS	137	Standard query response 0xdd3 AAAA dyna.wikimedia.org AAAA 2a02:ec80:...

Domain Name System (response)
Transaction ID: 0xdd3
Flags: 0x0100 Standard query response, No error
Questions: 1
Answer RRs: 1
Authority RRs: 0
Additional RRs: 1
Queries
dyna.wikimedia.org: type AAAA, class IN
Name: dyna.wikimedia.org
[Name Length: 18]
[Label Count: 3]
Type: AAAA (28) (IP6 Address)
Class: IN (0x0001)
Answers
dyna.wikimedia.org: type AAAA, class IN, addr 2a02:ec80:700:ed1a::1
Additional records

Domain Name System: Protocol | Packets: 43 · Displayed: 4 (9.3%) · Dropped: 0 (0.0%) | Profile: Default

O primeiro pacote mostra a solicitação do Ip do www.wikipedia.org pelo Type A o que mostra ser uma solicitação de IPV4. O retorno do servidor DNS mostra no Answers a resposta que o Ip versão 4 do www.wikipedia.org é 195.200.68.224. Logo em baixo é solicitado o IPV6 do wikipedia pelo Type AAAA, que é enviado pelo servidor DNS logo em seguida.

Exercício 8

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icmipv6

No.	Time	Source	Destination	Protocol	Length	Info
15	3.804330753	fe80::1	2804:10dc:f0f2:e100...	ICMPv6	86	Neighbor Solicitation for 2804:10dc:f0f2:e100:9621:7aec:c3e6:1c1b from...
16	3.804403597	2804:10dc:f0f2:e100...	fe80::1	ICMPv6	78	Neighbor Advertisement 2804:10dc:f0f2:e100:9621:7aec:c3e6:1c1b (sol)
21	4.095823393	2804:10dc:f0f2:e100...	2001:4860:482a:7700...	ICMPv6	118	Echo (ping) request id=0x9841, seq=1, hop limit=64 (reply in 22)
22	4.108261890	2001:4860:482a:7700...	2804:10dc:f0f2:e100...	ICMPv6	118	Echo (ping) reply id=0x9841, seq=1, hop limit=116 (request in 21)
27	5.097140974	2804:10dc:f0f2:e100...	2001:4860:482a:7700...	ICMPv6	118	Echo (ping) request id=0x9841, seq=2, hop limit=64 (reply in 28)
28	5.113054676	2001:4860:482a:7700...	2804:10dc:f0f2:e100...	ICMPv6	118	Echo (ping) reply id=0x9841, seq=2, hop limit=116 (request in 27)
30	6.098246657	2804:10dc:f0f2:e100...	2001:4860:482a:7700...	ICMPv6	118	Echo (ping) request id=0x9841, seq=3, hop limit=64 (reply in 31)
31	6.113735449	2001:4860:482a:7700...	2804:10dc:f0f2:e100...	ICMPv6	118	Echo (ping) reply id=0x9841, seq=3, hop limit=116 (request in 30)
34	7.099883758	2804:10dc:f0f2:e100...	2001:4860:482a:7700...	ICMPv6	118	Echo (ping) request id=0x9841, seq=4, hop limit=64 (reply in 35)
35	7.112755966	2001:4860:482a:7700...	2804:10dc:f0f2:e100...	ICMPv6	118	Echo (ping) reply id=0x9841, seq=4, hop limit=116 (request in 34)

Frame 21: 118 bytes on wire (944 bits), 118 bytes captured (944 bits) on
Ethernet II, Src: ChuangweiRbg_e0:9e:77 (60:42:7f:e0:9e:77), Dst: zte_21:
Internet Protocol Version 6, Src: 2804:10dc:f0f2:e100:9621:7aec:c3e6:1c1b
Internet Control Message Protocol v6
Type: Echo (ping) request (128)
Code: 0
Checksum: 0x4b6f [correct]
[Checksum Status: Good]
Identifier: 0x9841
Sequence: 1
[Response In: 22]
Timestamp from Echo data: Jun 6, 2026 02:26:22.358162000 -03
[Timestamp from Echo data (relative): 0.000018503 seconds]
Data (40 bytes)

Internet Control Message Protocol v6: Protocol | Packets: 71 · Displayed: 10 (14.1%) · Dropped: 0 (0.0%) | Profile: Default

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icmipv6

No.	Time	Source	Destination	Protocol	Length	Info
15	3.804330753	fe80::1	2804:10dc:f0f2:e100...	ICMPv6	86	Neighbor Solicitation for 2804:10dc:f0f2:e100:9621:7aec:c3e6:1c1b from...
16	3.804403597	2804:10dc:f0f2:e100...	fe80::1	ICMPv6	78	Neighbor Advertisement 2804:10dc:f0f2:e100:9621:7aec:c3e6:1c1b (sol)
21	4.095823393	2804:10dc:f0f2:e100...	2001:4860:482a:7700...	ICMPv6	118	Echo (ping) request id=0x9841, seq=1, hop limit=64 (reply in 22)
22	4.108261890	2001:4860:482a:7700...	2804:10dc:f0f2:e100...	ICMPv6	118	Echo (ping) reply id=0x9841, seq=1, hop limit=116 (request in 21)
27	5.097140974	2804:10dc:f0f2:e100...	2001:4860:482a:7700...	ICMPv6	118	Echo (ping) request id=0x9841, seq=2, hop limit=64 (reply in 28)
28	5.113054676	2001:4860:482a:7700...	2804:10dc:f0f2:e100...	ICMPv6	118	Echo (ping) reply id=0x9841, seq=2, hop limit=116 (request in 27)
30	6.098246657	2804:10dc:f0f2:e100...	2001:4860:482a:7700...	ICMPv6	118	Echo (ping) request id=0x9841, seq=3, hop limit=64 (reply in 31)
31	6.113735449	2001:4860:482a:7700...	2804:10dc:f0f2:e100...	ICMPv6	118	Echo (ping) reply id=0x9841, seq=3, hop limit=116 (request in 30)
34	7.099883758	2804:10dc:f0f2:e100...	2001:4860:482a:7700...	ICMPv6	118	Echo (ping) request id=0x9841, seq=4, hop limit=64 (reply in 35)
35	7.112755966	2001:4860:482a:7700...	2804:10dc:f0f2:e100...	ICMPv6	118	Echo (ping) reply id=0x9841, seq=4, hop limit=116 (request in 34)

Frame 22: 118 bytes on wire (944 bits), 118 bytes captured (944 bits) on
Ethernet II, Src: zte_21:09:8d (3c:f9:f0:21:09:8d), Dst: ChuangweiRbg_e0:
Internet Protocol Version 6, Src: 2001:4860:482a:7700::, Dst: 2804:10dc:f
Internet Control Message Protocol v6
Type: Echo (ping) reply (129)
Code: 0
Checksum: 0x4a6f [correct]
[Checksum Status: Good]
Identifier: 0x9841
Sequence: 1
[Response To: 21]
[Response Time: 12.438 ms]
Timestamp from Echo data: Jun 6, 2026 02:26:22.358162000 -03
[Timestamp from Echo data (relative): 0.012457000 seconds]
Data (40 bytes)

Internet Control Message Protocol v6: Protocol | Packets: 71 · Displayed: 10 (14.1%) · Dropped: 0 (0.0%) | Profile: Default

The screenshot shows a Wireshark packet capture of ICMPv6 traffic. The packet list on the left shows several packets, with packet 21 selected. The packet details pane on the right shows the structure of the selected packet, which is an Echo (ping) request. The packet bytes pane on the right shows the raw data of the packet.

No.	Time	Source	Destination	Protocol	Length	Info
15	3.804330753	fe80::1	2804:10dc:f0f2:e100::	ICMPv6	86	Neighbor Solicitation for 2804:10dc:f0f2:e100:9621:7aec:c3e6:1c1b from...
16	3.804403597	2804:10dc:f0f2:e100::	fe80::1	ICMPv6	78	Neighbor Advertisement 2804:10dc:f0f2:e100:9621:7aec:c3e6:1c1b (sol)
21	4.095823393	2804:10dc:f0f2:e100::	2001:4860:482a:7700::	ICMPv6	118	Echo (ping) request id=0x9841, seq=1, hop limit=64 (reply in 22)
22	4.108261890	2001:4860:482a:7700::	2804:10dc:f0f2:e100::	ICMPv6	118	Echo (ping) reply id=0x9841, seq=1, hop limit=116 (request in 21)
27	5.097140974	2804:10dc:f0f2:e100::	2001:4860:482a:7700::	ICMPv6	118	Echo (ping) request id=0x9841, seq=2, hop limit=64 (reply in 28)
28	5.113054676	2001:4860:482a:7700::	2804:10dc:f0f2:e100::	ICMPv6	118	Echo (ping) reply id=0x9841, seq=2, hop limit=116 (request in 27)
30	6.098246657	2804:10dc:f0f2:e100::	2001:4860:482a:7700::	ICMPv6	118	Echo (ping) request id=0x9841, seq=3, hop limit=64 (reply in 31)
31	6.113735449	2001:4860:482a:7700::	2804:10dc:f0f2:e100::	ICMPv6	118	Echo (ping) reply id=0x9841, seq=3, hop limit=116 (request in 30)
34	7.099883758	2804:10dc:f0f2:e100::	2001:4860:482a:7700::	ICMPv6	118	Echo (ping) request id=0x9841, seq=4, hop limit=64 (reply in 35)
35	7.112755966	2001:4860:482a:7700::	2804:10dc:f0f2:e100::	ICMPv6	118	Echo (ping) reply id=0x9841, seq=4, hop limit=116 (request in 34)

Frame 27: 118 bytes on wire (944 bits), 118 bytes captured (944 bits) on Ethernet II, Src: ChuangweiRbg_e0:9e:77 (60:42:7f:e0:9e:77), Dst: zte 21: Internet Protocol Version 6, Src: 2804:10dc:f0f2:e100:9621:7aec:c3e6:1c1b, Internet Control Message Protocol v6
 Type: Echo (ping) request (128)
 Code: 0
 Checksum: 0x4069 [correct]
 [Checksum Status: Good]
 Identifier: 0x9841
 Sequence: 2
 [Response In: 28]
 Timestamp from Echo data: Jun 6, 2026 02:26:23.359452000 -03
 [Timestamp from Echo data (relative): 0.000046084 seconds]
 Data (40 bytes)

The screenshot shows a Wireshark packet capture of ICMPv6 traffic. The packet list on the left shows several packets, with packet 28 selected. The packet details pane on the right shows the structure of the selected packet, which is an Echo (ping) reply. The packet bytes pane on the right shows the raw data of the packet.

No.	Time	Source	Destination	Protocol	Length	Info
15	3.804330753	fe80::1	2804:10dc:f0f2:e100::	ICMPv6	86	Neighbor Solicitation for 2804:10dc:f0f2:e100:9621:7aec:c3e6:1c1b from...
16	3.804403597	2804:10dc:f0f2:e100::	fe80::1	ICMPv6	78	Neighbor Advertisement 2804:10dc:f0f2:e100:9621:7aec:c3e6:1c1b (sol)
21	4.095823393	2804:10dc:f0f2:e100::	2001:4860:482a:7700::	ICMPv6	118	Echo (ping) request id=0x9841, seq=1, hop limit=64 (reply in 22)
22	4.108261890	2001:4860:482a:7700::	2804:10dc:f0f2:e100::	ICMPv6	118	Echo (ping) reply id=0x9841, seq=1, hop limit=116 (request in 21)
27	5.097140974	2804:10dc:f0f2:e100::	2001:4860:482a:7700::	ICMPv6	118	Echo (ping) request id=0x9841, seq=2, hop limit=64 (reply in 28)
28	5.113054676	2001:4860:482a:7700::	2804:10dc:f0f2:e100::	ICMPv6	118	Echo (ping) reply id=0x9841, seq=2, hop limit=116 (request in 27)
30	6.098246657	2804:10dc:f0f2:e100::	2001:4860:482a:7700::	ICMPv6	118	Echo (ping) request id=0x9841, seq=3, hop limit=64 (reply in 31)
31	6.113735449	2001:4860:482a:7700::	2804:10dc:f0f2:e100::	ICMPv6	118	Echo (ping) reply id=0x9841, seq=3, hop limit=116 (request in 30)
34	7.099883758	2804:10dc:f0f2:e100::	2001:4860:482a:7700::	ICMPv6	118	Echo (ping) request id=0x9841, seq=4, hop limit=64 (reply in 35)
35	7.112755966	2001:4860:482a:7700::	2804:10dc:f0f2:e100::	ICMPv6	118	Echo (ping) reply id=0x9841, seq=4, hop limit=116 (request in 34)

Frame 28: 118 bytes on wire (944 bits), 118 bytes captured (944 bits) on Ethernet II, Src: zte 21:69:8d (3c:f9:f0:21:69:8d), Dst: ChuangweiRbg_e0: Internet Protocol Version 6, Src: 2001:4860:482a:7700::, Dst: 2804:10dc:f0f2:e100:9621:7aec:c3e6:1c1b, Internet Control Message Protocol v6
 Type: Echo (ping) reply (129)
 Code: 0
 Checksum: 0x3f69 [correct]
 [Checksum Status: Good]
 Identifier: 0x9841
 Sequence: 2
 [Response To: 27]
 [Response Time: 15,914 ms]
 Timestamp from Echo data: Jun 6, 2026 02:26:23.359452000 -03
 [Timestamp from Echo data (relative): 0.015959786 seconds]
 Data (40 bytes)

Na linha 21, meu computador envia um pacote Echo request para testar a conexão o servidor da Google. Olhando os detalhes abaixo, o campo Type está com 128 e o Code com 0, mostrando que é um pedido de teste com o número de sequência 1. Na linha 22, o servidor da Google responde com um pacote Echo reply. Se abrirmos esse pacote, o campo Type muda para 129 (que significa resposta) e o número de sequência continua sendo 1, confirmando que o teste deu certo. Esse processo se repete nas próximas linhas.

Exercício 9

The first screenshot shows a DHCP traffic capture in Wireshark. The packet list shows two packets: a DHCP Request (337) and a DHCP ACK (360). The packet details pane shows the DHCP Request (337) with the following options:

- Option: (53) DHCP Message Type (Request)
- Option: (61) Client identifier

The packet bytes pane shows the raw data of the DHCP Request (337) and the DHCP ACK (360).

The second screenshot shows the same DHCP traffic capture, but with the DHCP ACK (360) expanded. The packet details pane shows the DHCP ACK (360) with the following options:

- Option: (53) DHCP Message Type (ACK)
- Option: (1) Subnet Mask (255.255.255.0)
- Option: (3) Router
- Option: (6) Domain Name Server
- Option: (54) DHCP Server Identifier (192.168.1.1)

The packet bytes pane shows the raw data of the DHCP ACK (360) and the DHCP Request (337).

No primeiro pacote ao expandir o Option [53] podemos ver que o meu computador faz uma request e não um discover mostrando que ao desconectar e conectar não houve o processo de descoberta ele apenas enviou uma requisição para reativar o IP antigo. No segundo pacote é devolvido um ack e definido a máscara e router.